MIL-A-17053E(SHIPS)
3 November 1967
SUPERSEDING
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2 May 1960
(See 6.4)

MILITARY SPECIFICATION

AUDIO ENTERTAINMENT SYSTEM EQUIPMENT, SHIPBOARD

1. SCOPE

1.1 Scope. This specification covers requirements for shipboard audio entertainment system equipment.

1.2 Classification. The entertainment system shall be of the following types, as specified (see 6.2):

Type I - Type I entertainment system shall be the entertainment system contained in a single enclosure. The type I entertainment system is designed primarily for surface ships (see 3.2.1).

Type II - Type II entertainment system shall be the entertainment system contained in three separate enclosures. The type II entertainment system was designed primarily for submarines (see 3.2.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-R-175 - Reels and Hubs for Magnetic Recording Tape, General Specification for.
W-T-70 - Tapes, Recording, Sound and Instrumentation, Magnetic Oxide Coated, General Specification for.

MILITARY

MIL-J-641 - Jacks, Telephone, General Specification for.
MIL-P-642 - Plugs, Telephone and Accessory Screws, General Specification for.
MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-H-5589 - Headphone, Extended Range, P/O H-3/APR-3.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for delivery of.
MIL-I-45208 - Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-275 - Printed Wiring for Electronic Equipment.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60606.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)
RS-238-A - Phonograph Stylus Tips

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, D. C. 20006.)

NATIONAL ASSOCIATION OF BROADCASTERS (NAB)
Standard for Phonograph Disc Recordings.
Standard for Magnetic Tape Recording and Reproducing (Reel to Reel).

(Application for copies should be addressed to the National Association of Broadcasters, 1771 N Street, Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction samples. Prior to production, a preproduction sample shall be examined and tested as specified in 4.2 (see 6.3).

3.2 Description. The audio entertainment system shall provide facilities for the simultaneous distribution of each of four audio channel outputs. Facilities shall be provided for recording and monitoring the system inputs and monitoring, muting, and attenuating the system outputs. The system shall contain two audio tape recorder-reproducers, an automatic record changer and all facilities necessary to provide the system features specified herein.

3.2.1 Type I and type II systems. The type I and type II entertainment systems are identical except for the rated output requirements for the power amplifier (see 3.5.1) and the mechanical requirements (see 3.10).

3.2.2 Equipment complement. Related ship entertainment system components not covered by this specification are broadcast radio receivers and loudspeakers.

3.3 General features. The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus - inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Ferrous alloys
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics

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Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Locking devices
Washers
Parts - electrical-general
Use of nonstandard parts
Requirements for semi-conductor devices (see 3.11.2)
Capacitors
Variable resistors
Transformers and inductors
Relays (see 3.11.1)
Electrical tapes
Switches
Indicator lights and lampholders
Fuses
Fuses and fuse clips (see 3.8.1)
Printed wiring
Enclosures - general (See 3.10.1.1 and 3.10.1.2)
Enclosure - accessibility
Enclosure - degree of (dripproof)
Enclosure - mounting (See 3.10.4)
Stiffening grooves
Cable entrance
Cable entrance plates (see 3.10.10.3)
Ventilation
Size (see 3.10.2)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Vibration and inclination (nonvital)
Features-electrical
Overload protection (see 3.5.4)
Primary power supply circuits (see 3.8)
Power supply tolerances (see 3.8)
Safety (personnel hazard)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Electrical connectors
Terminal boards and terminals (see 3.10.10.1)
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise (grade A)
Structureborne noise (type 3)
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Bill of materials
Interchangeability and standardization
Manuals (type II)
Repair parts (electronic)
Designation and marking (see 3.17)
Item names and nomenclature
Workmanship

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3.4 System operation.

3.4.1 Channel functions. The system shall provide for the simultaneous distribution of four individual audio program channels. The signal sources for channels 1 and 2 shall be the internal tape recorder-reproducer #1, the internal automatic record player, and the external radio #1 input source. The signal sources for channels 3 and 4 shall be the internal tape recorder-reproducer #2, the external auxiliary input source (see 3.4.4.1), and the external radio #2 input source. Each channel shall be provided with a selector switch for selecting any one of its signal sources for distribution to the channel output.

3.4.2 Recording functions. The system shall be provided with two identical 1/4 inch magnetic tape recorder-reproducers (see 3.6). Each one of the two recorder-reproducers shall be provided with facilities to:

(a) Record any one of the four channel signal sources (see 3.4.1), whether or not these signal sources are being distributed to the channel output at the time.
(b) Record only microphone input signals.
(c) Record any one of the four channels input sources and the microphone signals simultaneously.

3.4.3 Microphone functions. The system shall be provided with microphone recording (see 3.4.2) and microphone overriding facilities. The microphone overriding facility shall permit the microphone signal to be distributed over any channel output by overriding the signal or signals being distributed on that channel. Facilities shall be provided for overriding any one of the four channels individually or all of the four channels simultaneously. Adequate switching facilities shall be provided for selecting each microphone operational mode. A microphone level control having a range of at least 40 decibels (db) shall also be provided to adjust the microphone output level. The output level of each of the four channels shall not vary more than 1.0 db regardless as to whether the microphone is overriding an individual channel or all four channels simultaneously when tested in accordance with 4.5.6.

3.4.4 Inputs.

3.4.4.1 Operational signal inputs. The operational inputs to the system shall be provided as shown in table I.

Table I - Operational signal inputs.

Type of input	Impedance (Ohms) ± 10 percent	Input level (db)
Radio 1	600	0 dbm ± 20 db
Radio 2	600	0 dbm ± 20 db
Microphone	150	-50 dbm ± 15 db
Auxiliary	600	0 dbm ± 20 db

3.4.4.2 Control inputs.

3.4.4.2.1 Attenuating control input. The system shall be provided with facilities for attenuating the output levels of all four channels simultaneously. When the attenuating control input is short circuited the output level of each of the four channels shall be attenuated 13 ± 3 db. When the attenuating input is open circuited the output level of each of the four channels shall return to the original output level.

3.4.4.2.2 Muting control input. The system shall be provided with facilities for muting each of the four program channel outputs (see 3.4.5.1) when the muting control input is short circuited. An alarm indicator light located on the front panel of the equipment shall be lighted when the muting control input is short circuited and shall remain lighted until the muting has been manually defeated. An alarm switch shall be provided to manually defeat the muting facility and restore the system to normal operation.

3.4.5 Outputs.

3.4.5.1 Program channel outputs. A separate output shall be provided for distributing the output signals from each of the four program channels.

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3.4.5.2 Headset monitor output. A headset monitor output shall be provided on the front of the equipment for individually monitoring each of the four channel outputs, the record player output, each of the two recorder-reproducer output, and the auxiliary input. The headset monitor output shall have an impedance of 600 ± 60 ohms and a rated output level of 0 decibels relative to one milliwatt (dbm). A headset monitor output level control having a range of at least 40 db shall be provided to adjust the output level to the headset.

3.5 System characteristics.

3.5.1 Power amplifier. A power amplifier shall be provided for each of the four program channels. An output transformer shall be provided for each power amplifier.

3.5.1.1 Rated output.

3.5.1.1.1 Type I system. Each type I system power amplifier shall provide a 70 volt root mean square (rms) output into an output impedance of 90 ± 10 ohms.

3.5.1.1.2 Type II system. Each type II system power amplifier shall provide a 70 volt rms output into an output impedance of 240 ± 20 ohms.

3.5.2 Output level control. An output level control having a range of at least 40 db shall be provided for each power amplifier. A volume level (VU) meter with a selector switch shall be provided as an output level indicator for each of the four program channels. Zero db on the meter shall be 6 db below rated output level (see 3.5.1.1).

3.5.3 Tone controls. Each one of the four channels shall be provided with a single bass and a single treble control. The bass control shall adjust the 100 Hertz (Hz) output relative to the 1000 Hz output as specified in table II. The treble control shall adjust the 7,500 Hz output relative to the 1000 Hz output as specified in table II. The tone controls shall be marked "FLAT" at the midpoint of rotation, "MAX" at the maximum clockwise rotation, and "MIN" at the maximum counterclockwise rotation. The frequency response of the bass and treble controls in the "FLAT, MAX," and "MIN" positions shall be as shown in table II.

Table II - Bass and treble control response.

Control position	100 Hz response relative to the 1000 Hz response	7500 Hz response relative to the 1000 Hz response
FLAT	± 2 db	± 2 db
MAX	+ 12 ± 2 db	+ 12 ± 2 db
MIN	- 12 ± 2 db	- 12 ± 2 db

3.5.4 Overload protection. Overload protection shall be provided for each power amplifier (see 4.5.5).

3.6 Tape recorder-reproducer. The tape recorder-reproducer shall be provided with facilities for simultaneously recording and reproducing 1/4 track monaural magnetic tapes, reproducing 1/4 track stereo tapes monaurally, and reproducing 1/2 track monaural tapes (see NAB Standard for Magnetic Tape Recording and Reproducing). The recorder-reproducer shall contain all the mechanical components necessary for transporting tape and all the electronic components necessary for amplification and equalization during the record and reproduce modes. All mode controls shall be located on the front panel of the recorder-reproducer.

3.6.1 Recording and reproducing medium. The recording and reproducing medium shall be 1/4 inch wide polyester base magnetic tape in accordance with W-T-70.

3.6.2 Tape reel sizes. The recorder-reproducer shall be capable of operating with reels having either a 5/16 inch or 3 inch center hole diameter and a flange diameter up to and including 10-1/2 inches in accordance with W-R-175. Retainers for the tape reels shall be provided on each reel shaft.

3.6.3 Track dimensions. The track dimensions for two track and four track recordings shall be in accordance with the NAB Standard for Magnetic Tape Recording and Reproducing (Reel to Reel).

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3.6.4 Tape speeds. The tape shall move at linear speeds of 3.75 ± 0.05 and 7.5 ± 0.1 inches per second (i.p.s.) while recording and reproducing (see 3.8). A speed control switch shall be provided to select the two speeds. Selection of the equalization shall be made automatically when the speed control is operated.

3.6.5 Erasing. The equipment shall provide a means of erasing the recording by a high frequency erase method. Erasure shall automatically occur during the recording process such that the tape is erased prior to reaching the recording head.

3.6.6 Record mode. A control shall be provided to place the recorder-reproducer in the record mode. Facilities shall also be provided to prevent accidental erasure of tape by means of a suitable interlock. Indication that the recorder-reproducer is recording shall be provided on the front panel by a light.

3.6.6.1 Record level control. A record level control shall be provided on the front panel to adjust the level in the record mode.

3.6.7 Reproduce mode. A control shall be provided to place the recorder-reproducer in the reproduce mode.

3.6.8 Fast forward and rewind modes. The recorder-reproducer shall be provided with facilities for fast forward and rewind operation. Tape speed shall be at an average speed of at least 300 i.p.s.

3.6.9 Stop mode. A control shall be provided to stop the tape motion from any previously selected mode.

3.6.10 Power "ON-OFF" switch. A power switch shall be provided to disconnect the recorder-reproducer from the power supply line when in the "OFF" position. All rotating rubber parts in the recorder-reproducer shall be automatically disengaged when the power switch is in the "OFF" position.

3.6.11 Safety switch. A safety switch shall be provided to stop the tape transport in case of tape breakage or end of reel.

3.6.12 Tape path. An outline of the threading path shall be provided on the front panel of the tape deck.

3.6.13 Tape counter. A 3-digit mechanical counter shall be provided on the transport to give a measure of the tape used.

3.6.14 Optimum record level. An optimum record level shall be established by the contractor. Optimum record level shall be defined as the recording head current at 1000 Hz at which the recorder will meet all other recorder-reproducer requirements of this specification.

3.6.15 Equalization. Equalization shall be provided in the equipment. The equalization shall be in accordance with the NAB standard equalization curve for 3-3/4 and 7-1/2 i.p.s. tape speed.

3.6.16 Channel isolation. Channel isolation shall be at least minus 40 db when measured in accordance with 4.5.8.3.

3.6.17 Flutter. Flutter at a tape speed of 3.75 i.p.s. shall not exceed 0.3 percent r.m.s. when tested in accordance with 4.5.8.2.

3.7 Automatic record player. The automatic record player shall be capable of reproducing from one to ten records having a nominal spindle hole diameter of 1/4 inch or 1-1/2 inches, and an outside diameter of 7, 10 or 12 inches. The automatic record player shall also be capable of manual reproduction of records.

3.7.1 Automatic reproduction (single cycle). When a single record is placed on the recorder player spindle (see 3.12(g)) and the mode selection switch (see 3.7.12) is placed in the "REJECT" position and released, the record player mechanism shall place the pickup arm at the start of the record. During the reproduction of the record, the record player mechanism shall not in any way affect the freedom of either the lateral or vertical motion of the pickup arm. When the pickup reaches the record's tripping groove and operates the tripping mechanism (see 3.7.3) the pickup arm shall be returned to the arm rest position (see 3.7.7) and the mode selection switch shall go to the "OFF" mode (see 3.7.12.1).

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3.7.2 Automatic reproduction (multiple cycle). When more than one record is placed on the record player spindle (see 3.12(g)) and the mode switch is placed in the "REJECT" position and released the record player shall drop the first record onto the turntable and place the pickup arm at the start of the record. During the reproduction of the first record, the record player mechanism shall not in any way affect the freedom of either the lateral or vertical motion of the pickup arm. When the pickup reaches the record's tripping grooves and operates the tripping mechanism the pickup arm shall be returned to the arm rest position. Then the next record on the record player spindle shall be dropped on to the turntable and the reproduction cycle repeated. When the pickup reaches the tripping grooves of the last record to be reproduced the pickup arm shall be returned to the arm rest position and the mode selection switch shall go to the "OFF" mode.

3.7.3 Trip mechanism force. When the pickup reaches the tripping grooves of the record, the lateral force required to trip the record player mechanism shall not exceed 1.0 gram when tested in accordance with 4.5.7.3.

3.7.4 Turntable platen. The record player shall employ a minimum 10-inch diameter turntable with interchangeable center facilities (see 3.12(g)and (h)) for manual and automatic reproduction of records. The turntable platen shall be constructed from a solid nonferrous metal weighing at least 6.5 pounds and shall be covered by a vibration absorbent pad at least 1/8 inch thick. Provision shall be made for locking the turntable to prohibit unintentional removal of the turntable from the drive shaft.

3.7.5 Drive mechanism. The turntable drive mechanism shall employ a rim drive system. Drive idlers shall be disengaged from associated driving or driven members when the mode selection switch (see 3.7.12) is in the "OFF" position.

3.7.6 Pickup arm. The pickup arm shall be a massed balanced type. The simultaneous condition of pitch and roll shall not affect the operation of the record player in any manner when tested in accordance with 4.5.7.10.

3.7.7 Pickup arm clamp. A pickup arm clamp shall be provided to positively secure the pickup arm assembly in the arm rest position when the record player is not in use. The automatic record player shall not be damaged if the record player reproduction cycle is initiated (see 3.7.12) when the pickup arm is secured by the clamp.

3.7.8 Pickup cartridge. The pickup cartridge shall be a magnetic type stereo cartridge wired for monaural reproduction. The cartridge holder shall be designed for easy cartridge mounting and shall lock the cartridge rigidly in place to act as an integral part of the pickup arm. The design of the cartridge shall be such that the stylus can be easily replaced. Dropping or skidding on the record surface or turntable pad shall not result in damage to the cartridge or stylus.

3.7.9 Pickup cartridge stylus. The pickup stylus shall have a 0.7 mil (0.0007 inch) tip radius and shall be constructed from a whole diamond. The pickup cartridge stylus shall have a 15-degree vertical tracking angle in accordance with the EIA standards RS-238-A.

3.7.10 Record size selector switch. The record player shall be provided with a record size selector switch to select record sizes of 7, 10 and 12 inches. The record size selector switch shall be located on the record player mechanism.

3.7.11 Turntable speed. The record player shall be provided with a turntable speed control to select turntable speeds of 16-2/3, 33-1/3, 45 and 78 r.p.m. (revolutions per minute) + 0.5 percent (see 3.8) when tested in accordance with 4.5.7.4. The turntable speed control shall be marked to indicate the nominal speed of the turntable in r.p.m.

3.7.12 Mode selection switch. A four position mode selection switch shall be provided to place the record player in the REJECT, ON, OFF and MANUAL modes. The mode selection switch shall be located beyond the outer edge of the turntable.

3.7.12.1 "OFF" mode. When the mode selection switch in the "OFF" position the record player turntable shall not rotate.

3.7.12.2 "ON" mode. When the mode selection switch is in the "ON" position the record player turntable shall rotate at the selected speed.

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3.7.12.3 "REJECT" mode. When the mode selection switch is held in the "REJECT" position and released (the switch shall automatically return to the "ON" position when released) the record player shall function as follows:

(a) If the record player is in the "OFF" mode, the reproduction cycle shall be initiated (see 3.7.1 and 3.7.2).
(b) If a record is being reproduced and there are other records on the record player spindle (see 3.12(g)), reproduction of the record being reproduced shall be discontinued and a reproduction cycle initiated for the next record on the record player spindle.
(c) If a record is being reproduced and there are no other records on the record player spindle, reproduction of the record being reproduced shall be discontinued and the mode selection switch shall be placed in the "OFF" mode.

3.7.12.4 "MANUAL" mode. When the mode selection switch is placed in the "MANUAL" position the record player turntable shall rotate at the selected speed. Automatic reproduction (see 3.7.1 and 3.7.2) shall not be possible in the "MANUAL" mode.

3.7.13 Stylus force gauge. A stylus force gauge with indicating scale shall be provided on the pickup arm to indicate the stylus force on a record. The gauge shall be calibrated from 0 to 7 grams in 0.5 gram increments and shall be accurate within 0.1 gram when measured in accordance with 4.5.7.2.

3.7.14 Pickup amplifier. The pickup amplifier shall provide record player playback equalization in accordance with the NAB standard on Phonograph Disc Recordings.

3.7.15 Pickup arm resonance. Pickup arm and cartridge assembly resonance shall be less than 15 Hz when tested in accordance with 4.5.7.6.

3.7.16 Flutter. The flutter (wow) of the turntable shall not exceed 0.2 percent r.m.s. at 33-1/3 r.p.m. when tested in accordance with 4.5.7.7.

3.7.17 Vibration isolation. The turntable, the turntable motor, and record changer mechanism shall be resiliently mounted with suitable damping provided. The resonant frequencies of the mechanism or its mount shall be sufficiently displaced from the other resonant frequencies (such as pickup arm and motor speeds) to keep rumble at least 30 db below rated output when measured in accordance with 4.5.7.8.

3.7.18 Tracking error. The pickup arm shall be located so that the tracking error from 100 to 7500 Hz shall not produce more than 1 percent second harmonic distortion when tested in accordance with 4.5.7.5.

3.8 Power supply. The equipment shall be designed to operate from 115 volts, 60 Hz, single phase electrical power in accordance with MIL-I-983 except for the recorder-reproducer tape speeds (see 3.6.4) and automatic record player turntable speeds (see 3.7.11).

3.8.1 Fuses and fuse clips. Fuses and fuse clips shall provide individual fusing for both sides of the line for each system component requiring primary power.

3.8.2 Power switch. A power switch shall be provided which will disconnect both sides of the power supply lines from all the equipment components. A pilot light shall also be provided and it shall remain lighted when the power switch is in the "ON" position.

3.9 System performance requirements.

3.9.1 Frequency response. The frequency response of the system shall be 100 to 7500 Hz $\pm$ 2 db when tested in accordance with 4.5.2.

3.9.2 Harmonic distortion. The total harmonic distortion in the system output shall not exceed 4 percent, when tested in accordance with 4.5.5.

3.9.3 Regulation. Variations in loading on power amplifiers between full load (see 3.5.1.1) and no load shall not cause the output voltage to vary by more than 3 db at any frequency from 100 to 7500 Hz when tested as specified in 4.5.9.

3.9.4 Signal-to-noise.

3.9.4.1 Microphone function. The signal-to-noise ratio for the microphone function shall be at least 50 db when tested in accordance with 4.5.4.1.

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3.9.4.2 Tape recorder-reproducer and record player. The signal-to-noise ratio for the tape recorder-reproducer and the record player shall be at least 35 db when tested in accordance with 4.5.4.2 and 4.5.4.3.

3.9.5 Crosstalk. The crosstalk between any two channels in the system shall be not greater than 35 db when tested in accordance with 4.5.6.

3.10 Mechanical requirements.

3.10.1 System configurations.

3.10.1.1 Type I system. The type I system shall consist of packaging all the components in a single enclosure.

3.10.1.2 Type II system. The type II system shall consist of three separate units (see 3.10.1.2.1 through 3.10.1.2.3).

3.10.1.2.1 Unit 1. Unit 1 shall contain the system control panel and the four channel power amplifiers in a single enclosure. All external input and output connections (see 3.10.10) shall be to unit 1.

3.10.1.2.2 Unit 2. Unit 2 shall contain the two recorder-reproducers in a single enclosure.

3.10.1.2.3 Unit 3. Unit 3 shall contain the record player in a single enclosure.

3.10.1.2.4 Interconnection. The inter unit cabling connections shall be made to terminal boards. Cable entrance plate requirements of 3.10.10.3 apply to the individual units.

3.10.2 Dimensions. The system shall not exceed the dimensions shown in table III. All dimensions shall be measured to include shock mounts (see 3.10.4), projections and controls on the body of the equipment. The system shall be capable of passage through a circular hatch 25 inches in diameter.

Table III - Dimensions (inches)

	Height	Width	Depth
Type I	55	22	20
Type II			
Unit 1	12	22	13
Unit 2	30	22	20
Unit 3	14	15	15

3.10.3 Weight. The weight of the type I system shall not exceed 250 pounds. The combined weight of all three units of the type II system shall not exceed 225 pounds.

3.10.4 Mounting. The type I system and the units of the type II system shall be designed for shelf mounting. Shock insulators may be used if required. If shock mounts are used, a grounding strap shall be provided to insure a good conducting path between the enclosure and the mounting base.

3.10.5 Tape recorder-reproducers. The supporting panel for each recorder-reproducer shall be fastened to enclosure flanges by means of easily removable screws; all electrical connections to the mechanism shall be made with separable connectors, to permit easy removal of the mechanism from the front for servicing.

3.10.6 Record player. The automatic record player (see 3.7) shall be installed in a drawer with facilities provided for positively securing the drawer in either an open or closed position. The record player shall be capable of operation with the drawer either open or closed.

3.10.7 Power amplifier. The power amplifier chassis shall be plug-in type supported by slides inside the enclosure. The chassis shall be provided with adequate hand grips to facilitate removing and handling during service operations. The chassis shall be held in

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their guides by locking type fasteners requiring no tools for operation. Each chassis shall contain its own power supply components, and no exterior connections to the power amplifiers shall be required other than those for input and output circuits and for the 115 volt primary power circuits.

3.10.8 Controls. All controls shall be located on the front panel of the equipment.

3.10.9 Modular construction. Modular construction shall be used where possible. Amplifiers and oscillators using modular construction shall have components mounted on circuit boards with type V lead terminations in accordance with MIL-STD-275.

3.10.10 Input-output connections.

3.10.10.1 Terminal boards. Separate terminal board connections shall be provided for both radio inputs (see 3.4.1), both control inputs (see 3.4.4.2), and each of the four program channel outputs.

3.10.10.2 Jacks. Individual jacks type JJ-089 in accordance with MIL-J-641 shall be provided on the front of the equipment for the microphone input, the auxiliary input and the headset monitor output.

3.10.10.3 Cable entrance plates. Removable cable entrance plates shall be provided so that cables may enter the enclosure from the top and back.

3.11 Electrical components.

3.11.1 Relays. Relays shall be provided with removable dustproof covers.

3.11.2 Semiconductor devices. The equipment shall employ transistors and other simiconductor devices exclusively where active elements are required.

3.11.3 Motors. All motors used in the equipment shall be life lubricated and shall not require any additional lubrication.

3.11.4 Level control linearity. For equal angles of rotation, the attenuation provided by the level controls shall be essentially equal in decibels (db) throughout at least 80 percent of control rotation.

3.12 Accessory kit. Each equipment shall be provided with a kit containing the following items:

(a) One microphone, Shure model 51, or equal. The microphone case shall be designed for desk mounting, and a microphone desk stand shall be supplied. The microphone shall also be supplied with a 15-foot shield cable terminated with a type PJ-055B plug in accordance with MIL-P-642.
(b) One headset in accordance with MIL-H-5589. The headset shall be supplied with a 6-foot shielded cable terminated with a type PJ-055B plug in accordance with MIL-P-642.
(c) One 10-1/2-inch reel of unrecorded 1/4-inch magnetic tape in accordance with W-R-175 and W-T-70.
(d) Two 10-1/2-inch reels and two 7-inch reels for 1/4-inch magnetic tape in accordance with W-R-175.
(e) One commercial grade splicer for 1/4-inch magnetic tape.
(f) One roll of commercial grade splicing tape.
(g) One centering spindle for automatic reproduction of ten 45 r.p.m. records and one centering spindle for automatic reproduction of ten 33-1/3 records.
(h) One centering spindle for manual reproduction of 33-1/3 r.p.m. records and one centering spindle for manual reproduction of 45 r.p.m. records.

3.13 Temperature and humidity. The equipment shall show no degradation in electrical or mechanical performance after being tested in accordance with 4.5.14.

3.14 Accelerated life. The equipment shall be capable of 250 hours continuous operation and show no degradation in electrical or mechanical performance after being tested in conformance with 4.5.15.

3.15 Shock. The equipment shall meet the shock requirements for grade B equipments when tested as type A unit in accordance with MIL-S-901.

3.16 Design for servicing. The equipment shall be designed so that servicing of all parts can be quickly and easily performed.

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3.17 Designation and marking. In addition to the designation and marking requirements of MIL-I-983, all controls shall be permanently marked to indicate the function performed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier shall provide and maintain an inspection system acceptable to the Government for the supplies covered by this specification. The system of inspection shall be in accordance with MIL-I-45208.

4.2 Preproduction inspection. Preproduction inspection shall consist of the examination and tests listed in table IV, which shall be performed in general in the order listed.

Table IV - Preproduction inspection

Examination and tests	Requirement paragraph	Examination and test reference
General examination	3.3	MIL-I-983
Operating	---	4.5.1
Supply line voltage and frequency variation	3.3	MIL-I-983
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Frequency response	3.9.1	4.5.2
Tone controls	3.5.3	4.5.3
Signal-to-noise	3.9.4	4.5.4
Crosstalk and microphone level	3.4.3 and 3.9.5	4.5.6
Overload protection and harmonic distortion	3.5.4 and 3.9.2	4.5.5
Turntable platen	3.7.4	4.5.7.1
Stylus force gauge	3.7.13	4.5.7.2
Trip mechanism force	3.7.3	4.5.7.3
Turntable speed	3.7.11	4.5.7.4
Tracking error	3.7.18	4.5.7.5
Pickup arm resonance	3.7.15	4.5.7.6
Record player flutter	3.7.16	4.5.7.7
Vibration isolation	3.7.17	4.5.7.8
Pitch and roll	3.7.6	4.5.7.9
Pickup arm clamp	3.7.7	4.5.7.10
Tape speeds	3.6.4 and 3.6.8	4.5.8.1
Recorder-reproducer flutter	3.6.17	4.5.8.2
Channel isolation	3.6.16	4.5.8.3
Regulation	3.9.3	4.5.9
Input and output impedance	3.4.4.1, 3.4.5.2 and 3.5.1.1	4.5.10
Input level range	3.4.4.1	4.5.11
Level control linearity	3.11.4	4.5.12
Mechanical requirements	3.10	4.5.13
Airborne noise	3.3	MIL-I-983
Structureborne noise	3.3	MIL-I-983
Dielectric strength and insulation resistance clearances	3.3	MIL-I-983
Temperature and humidity	3.13	4.5.14
Accelerated life	3.14	4.5.15
Shock	3.15	MIL-S-901
Vibration	3.3	MIL-I-983

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4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Group A examination and tests. All equipments covered on the contract or order shall be subjected to the group A examination and tests listed in table V. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination and test shall be counted as a defect, and the equipment shall not be offered for delivery.

Table V - Quality conformance inspection

Examination and tests	Requirement paragraph	Examination test and reference
Group A		
General examination	3.3	MIL-I-983
Operating	---	4.5.1
Group B		
Frequency response	3.9.1	4.5.2
Tone controls	3.5.3	4.5.3
Signal-to-noise	3.9.4	4.5.4
Overload protection and harmonic distortion	3.5.4 and 3.9.2	4.5.5
Crosstalk and microphone level	3.4.3 and 3.9.5	4.5.6
Channel isolation	3.6.16	4.5.8.3
Regulation	3.9.3	4.5.9
Input level range	3.4.4.1	4.5.11
Record player flutter	3.7.16	4.5.7.7
Recorder-reproducer flutter	3.6.17	4.5.8.2
Level control linearity	3.11.4	4.5.12
Group C		
Supply line voltage and frequency variation	3.3	MIL-I-983
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Dielectric strength and isulation resistance clearances	3.3	MIL-I-983
Airborne noise	3.3	MIL-I-983
Structureborne noise	3.3	MIL-I-983
Insulation resistance clearances	3.3	MIL-I-983
Mechanical requirements	3.10	4.5.13
Temperature and humidity	3.13	4.5.14
Accelerated life	3.14	4.5.15
Shock	3.15	MIL-S-901
Vibration	3.3	MIL-I-983
Input and output impedance	3.4.4.1, 3.4.5.2 and 3.5.1.1	4.5.10
Turntable platen	3.7.4	4.5.7.1
Stylus force gauge	3.7.13	4.5.7.2
Trip mechanism force	3.7.3	4.5.7.3
Turntable speed	3.7.11	4.5.7.4
Tracking error	3.7.18	4.5.7.5
Pickup arm resonance	3.7.15	4.5.7.6
Vibration isolation	3.7.17	4.5.7.8
Pitch and roll	3.7.6	4.5.7.9
Pickup arm clamp	3.7.7	4.5.7.10
Tape speeds	3.6.4 and 3.6.8	4.5.8.1

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4.3.3 Sampling for group B tests. A sample of equipments shall be selected from each lot in accordance with table VI and subjected to the group B tests listed in table V with lot acceptance based on the sampling requirements of MIL-STD-105. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect, and the equipment shall not be offered for delivery. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

Table VI - Sampling for group B tests

Number of equipments in lot	Number of equipments in samples	Number of equipments nonconforming on group B tests	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.3.4 Sampling for group C tests. Group C tests will be required by the command or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in table V. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.4 Quality conformance inspection. The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests listed in table V. Examination and tests shall be performed, in general, in the order listed.

4.5 Test procedures. All tests shall be performed with 115 volts, 60 Hz, single phase, being supplied to the equipment. CBS Laboratories (Stamford, Connecticut) technical series test records are specified herein for use in the record player tests; however, it is recognized that other test records may also be satisfactory. Any alternate test record shall not be employed until prior approval is obtained from the Naval Ship Engineering Center. All record player tests shall be performed at a stylus force optimum for the cartridge.

4.5.1 Operating test. The operating test shall consist of the following:

(a) Two 12-inch commercial phonograph records shall be reproduced at 33-1/3 r.p.m. through the channel 1 output to determine conformance with 3.7.2.
(b) Two 7-inch commercial phonograph records shall be reproduced at 45 r.p.m. through the channel 2 output to determine conformance with 3.7.2.
(c) A signal shall be applied to each of the four channel inputs. The microphone supplied with the equipment shall be used to override the output signals from the four channels to determine conformance with 3.4.3.
(d) A commercial 1/4-track stereo tape shall be reproduced on recorder-reproducer #1. The output signals from recorder-reproducer #1 shall be simultaneously recorded on recorder-reproducer #2, and reproduced through the channel 1 output. The signals recorded on recorder-reproducer #2 shall be reproduced through channel #4 output.
(e) A signal shall be applied to each of the four channel inputs. The output signals from each of the four channels shall be attenuated to determine conformance with 3.4.4.2.1 and then muted to determine conformance with 3.4.4.2.2.
(f) All electrical and mechanical controls shall be actuated to determine conformance with the requirements of this specification.

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4.5.2 Frequency response. The outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). All tone controls shall be placed in the "FLAT" position. A signal generator having a constant output voltage with frequency and not more than 0.3 percent harmonic distortion at any frequency within the test range of 100 to 7500 Hz shall be used as the input signal source.

4.5.2.1 Radio input. A 1000 Hz signal at minus 20 dbm shall be applied to the radio #1 input and the channel 1 output level control shall be adusted to give rated output level across the channel 1 output resistive load. Signal frequencies of 100, 200, 400, 700, 2000, 4000, 6000 and 7500 Hz at minus 20 dbm shall be applied to the radio #1 input and the output at these frequencies shall be measured across the output resistive load with the channel 1 output level control set at the same level as when rated output level at 1000 Hz was attained. These output levels shall be related in db to the 1000 Hz output level to determine conformance with 3.9.1. The test shall be repeated on channel 3 using the radio #2 input.

4.5.2.2 Recorder-reproducer. A 1000 Hz signal at minus 20 dbm shall be applied to the auxiliary input. The 1000 Hz signal shall be recorded on recorder-reproducer #1 at 3.75 i.p.s. with the record level control set for optimum record level. The 1000 Hz signal shall be reproduced and the channel 2 output level control adjusted to give rated output level across the channel 2 output resistive load. Frequencies of 100, 200, 400, 2000, 5000 and 7500 Hz at minus 20 dbm shall be applied to the auxiliary input, recorded, and reproduced with record and channel 2 output level controls set at the same level as when rated output level at 1000 Hz was attained. These output levels shall be related in db to the 1000 Hz level to determine conformance with 3.9.1. The test shall be repeated at 7.5 i.p.s. to determine conformance with 3.9.1. Both tests shall be repeated using recorder-reproducer #2 and channel 3.

4.5.2.3 Record player. A frequency test record, CBS Laboratories technical series test record BTR-150 shall be used as a signal source. The channel 2 selector switch shall be placed in the record player position. The 1000 Hz signal on side B, band 3 of the BTR-150 test record, shall be reproduced on the automatic record player and the channel 2 output level control shall be adjusted to give rated output across the channel 2 output resistive load. All test record frequencies on side B, band 3 from 100 to 8000 Hz shall be reproduced on the record player and the channel 2 output at these frequencies shall be measured across the output resistive load with the channel 2 output level control set at the same level as when the rated output level at 1000 Hz was attained. These output levels shall be related in db to the 1000 Hz output level to determine conformance with 3.9.1. The test shall be repeated using side B, band 4 of the BTR-150 test record to determine conformance with 3.9.1.

4.5.2.4 Microphone. The monitor headset output shall be terminated with a 600 ohm resistive load. A 1000 Hz signal of minus 65 dbm shall be applied to the microphone input. The microphone selector switch shall be placed in the channel 3 position and the headset monitor selector switch placed in the channel 3 position. The microphone and channel 3 output level controls shall be adjusted to give rated output across the channel 3 output resistive load. The headset monitor level control shall be adjusted to give 0 dbm at the headset monitor output. Signal frequencies of 100, 200, 700, 2000, 4000, 6000 and 7500 Hz at minus 65 dbm shall be applied to the microphone input and the output at these frequencies shall be measured across the channel 3 and headset monitor resistive loads with all level controls set at the level as when rated output level at 1000 Hz was attained. These output levels shall be related in db to the 1000 Hz output level to determine conformance with 3.9.1.

4.5.3 Tone controls. The outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). All tone controls shall be placed in the "FLAT" position. A 1000 Hz signal at minus 20 dbm shall be applied to the radio #1 input and the channel 1 output level control shall be adjusted to give 12 db below rated output across the channel 1 output resistive load. A 1000 Hz signal shall be removed, and a 100 Hz shall be applied to the radio #1 input. With the output level control set at the same level as when 12 db below rated output was attained, the bass control shall be placed in the "MIN" position and the 100 Hz response measured across the resistive load shall be related in db to the 1000 Hz response to determine conformance with 3.5.3. With the output level control set at the same level as when 12 db below rated output was attained, the bass control shall be placed in the "MAX" position and the 100 Hz response measured across the resistive load shall be related in db to the 1000 Hz response to determine conformance with 3.5.3. The test shall be repeated using the treble control and a 7500 Hz input signal.

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4.5.4 Signal to noise. The outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). The channel 1, channel 2, and auxiliary inputs shall be terminated with a 600 ohm resistive load and the microphone input shall be terminated with a 150 ohm resistive load. All tone controls shall be placed in the "FLAT" position. The recorder-reproducer and the record player shall be in operation during all noise measurements whether or not they are being used in the particular operational mode under test.

4.5.4.1 Microphone. The microphone mode selector switch (see 3.4.3) shall be placed in the position for overriding all of the four channels simultaneously. A 1000 Hz signal of minus 65 dbm shall be applied to the microphone input. The microphone and the four channel output level controls shall be adjusted to give rated output level across their respective resistive loads. The noise level of each channel amplifier shall be measured across each output resistive load with no signal applied to the microphone input and with all level controls set at the same level as when rated output level at 1000 Hz was attained. The noise level of each channel shall be expressed in db relative to rated output level to determine conformance with 3.9.4.1.

4.5.4.2 Recorder-reproducer. A 1000 Hz signal at a level of minus 20 dbm shall be applied to the auxiliary input. This signal shall be recorded at 7.5 i.p.s. on recorder-reproducer #1 with the record level control adjusted for optimum record level. The tape shall be reproduced through channel 1 with the channel 1 output level control adjusted to give rated output level across the channel 1 output resistive load. The recorded portion of the tape shall be erased on recorder-reproducer with the record level control set at optimum record level while recording with no input signals. The erased portion of the tape shall then be reproduced with the channel 1 output level control set at the same level as when the 1000 Hz signal was reproduced. The noise level shall be measured across the output resistive load and expressed in db relative to the 1000 Hz rated output level to determine conformance with 3.9.4.2.

4.5.4.3 Record player. A test record, CBS Laboratories technical series Test Record BTR-150 (side B, band 1) shall be reproduced on the record player through channel 2, with the channel 2 output level control adjusted to give rated output level across the channel 2 resistive load. Band 6, side A, of the BTR-150, test record shall be reproduced on the record player with the channel 2 output level control set at the same level as when band 1, side B of the BTR-150 test record was reproduced. The noise level shall be measured across the output resistive load and expressed in db relative to the rated output level to determine conformance with 3.9.4.2. The test shall be repreated using side B, band 2 and side A, band 6 of the BTR-150 test record to determine conformance with 3.9.4.2.

4.5.5 Overload protection and harmonic distortion. The outputs of each channel amplifier shall be terminated with its proper resistive load (see 3.5.1.1). All tone controls shall be placed in the "FLAT" position. A 1000 Hz signal of minus 65 dbm shall be applied to the microphone input. The microphone mode selector switch (see 3.4.3) shall be placed in the position for overriding all of the four channels simultaneously. Each channel output level control shall be adjusted to give rated output power across their respective output resistive loads. With the level controls in the same position as when rated output level was attained, the outputs of each channel shall be terminated with a resistive load less than or equal to 1.0 ohm for a period of 10 minutes. After the 10 minute period, the microphone input signal shall be removed and the outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). The distortion shall then be measured (see 4.5.5.1 through 4.5.5.3). A signal generator having a constant output voltage with frequency and not more than 0.3 percent harmonic distortion shall be used as the input signal source.

4.5.5.1 Recorder-reproducer. Input signals of minus 20 dbm shall be recorded on recorder-reproducer #1 at 7.5 i.p.s. through the radio #1 input for frequencies of 100, 1000, and 2500 Hz. Each of the signals shall be recorded at optimum record level. The tape shall then be reproduced through channel 1 with the channel 1 output level control adjusted to give rated output power across the resistive load. Distortion shall be measured to determine conformance to 3.9.2. The test shall be repeated on channel 3 using recorder-reproducer #2.

4.5.5.2 Record player. Spot frequencies of 100,1000, and 2000 Hz on the CBS Laboratories technical series Test Record BTR-150 (side B, band 1) shall be reproduced on the record player through channel 2 with the channel 2 output level control adjusted to give rated output level across the resistive load. Distortion shall be measured to determine conformance with 3.9.2.

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4.5.5.3 Microphone. The microphone mode selector switch (see 3.4.3) shall be placed in the position for overriding all of the four channels simultaneously. Input signals of minus 65 dbm for frequencies of 100, 1000, and 2500 Hz shall be applied to the microphone input. The microphone level control and the output level controls of each of the four channels shall be adjusted to give rated output level across their respective resistive loads. Distortion shall be measured at each channel output to determine conformance to 3.9.2. The test shall be repeated for an input signal of minus 35 dbm.

4.5.6 Crosstalk and microphone level. The outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). All tone controls shall be placed in the "FLAT" position. A 1000 Hz signal of plus 65 dbm shall be applied to the microphone input. The microphone selector switch shall be placed in the position for overriding all of the four channels simultaneously. Each channel output level control shall be adjusted to give rated output level across their respective output resistive loads. The microphone selector switch shall then be placed in the position for overriding channel 1 only. With all level controls set at the same level as when rated output level was attained, the output level of channel 1 shall be measured and related in db to rated output level to determine conformance with 3.4.3. The output levels of channels 2, 3, and 4 shall be related to db to the channel 1 output level at 1000 Hz to determine conformance with 3.9.5. The tests shall be repeated with the microphone selector switch overriding channels 2, 3, and 4 individually to determine conformance with 3.9.5 and 3.4.3.

4.5.7 Record player tests.

4.5.7.1 Turntable platen. The turntable platen shall be weighed to determine conformance with 3.7.4.

4.5.7.2 Stylus force gauge. The stylus force gauge scale shall be measured with a balanced scale to determine conformance with 3.7.13.

4.5.7.3 Trip mechanism force. The trip mechanism force required to trip the record changer mechanism shall be measured with a gram gauge to determine conformance with 3.7.3. The gram gauge shall be accurate within 0.1 gram.

4.5.7.4 Turntable speed. A stroboscope disc and stroboscope or test record shall be used to determine compliance with 3.7.11. The turntable speeds shall conform to 3.7.11 over the range of power supply voltage tolerances specified in MIL-I-983. Turntable speeds shall only be tested at a power supply frequency of 60 Hz.

4.5.7.5 Tracking error A test record, CBS Laboratories technical series test record BTR-150, side B, band 1 shall be used as a signal source. The distortion shall be measured at the pickup amplifier output to determine conformance with 3.7.18. The test shall be repeated using side B, band 2 of the BTR-150 test record as a signal source to determine conformance with 3.7.18.

4.5.7.6 Pickup arm resonance. A test record, CBS Laboratories technical series test record STR-100, side A, bands 4 and 5, shall be used to determine conformance with 3.7.15.

4.5.7.7 Flutter. A test record, CBS Laboratories technical series test record STR-150 side A, band 5 shall be used with a suitable flutter and wow meter, to determine conformance with 3.7.16.

4.5.7.8 Vibration isolation. The channel 4 output shall be terminated with its proper resistive load (see 3.5.1). The channel 4 input selector switch shall be placed in the record player position and a test record, CBS Laboratories technical test record STR-120 shall be used as a signal source. Band 3 of the STR-120 test record shall be reproduced on the record player and the channel 4 output level control shall be adjusted to give rated output level across the output resistive load. Band 3 of the test record shall be removed and band 1 of the STR-120 test record shall be reproduced on the record player. With the channel 4 level control set in the same level as when rated output level was attained, the output level of band 1 of the STR-120 test record shall be measured and related in db to the output level of band 3 of the STR-120 test record to determine conformance with 3.7.17. The test shall be repeated relating bands 3 and 8, bands 2 and 1, bands 2 and 8, of the STR-120 test record to determine conformance with 3.7.17.

4.5.7.9 Pitch and roll. The equipment shall reproduce automatically two records at 33-1/3 r.p.m. and reproduce automatically two records at 45 r.p.m. while undergoing simultaneous pitch and roll as specified in 4.5.7.9.1 and 4.5.7.9.2 to determine conformance with 3.7.6.

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4.5.7.9.1 Roll. The roll (motion with respect to axis parallel to the pickup arm when in the arm rest position) shall be through an angle of 50 degrees (± 25 degree pitch) at the rate of 10 seconds per cycle with an acceleration varying with an approximate sine wave form.

4.5.7.9.2 Pitch. The pitch (motion with respect to axis perpendicular to the pickup arm when in the arm rest position) shall be through an angle of 16 degrees (± 8 degrees) at the rate of 15 seconds per cycle with an acceleration varying with an approximate sine wave form.

4.5.7.10 Pickup arm clamp. The record changer reproduction cycle shall be initiated (see 3.7.12.3) with the pickup arm secured by the pickup arm clamp. Then the operating test (see 4.5.1) shall be performed to determine conformance with 3.7.7. Flutter and tracking error tests shall also be performed to determine conformance with 3.7.16 and 3.7.18 respectively.

4.5.8 Recorder-reproducer tests. The tests specified in 4.5.8.1 through 4.5.8.3 shall be performed on each recorder-reproducer.

4.5.8.1 Tape speeds. Tape speeds shall be measured to determine conformance with 3.6.4 and 3.6.8.

4.5.8.2 Flutter. A 3000 Hz signal shall be recorded at optimum record level (see 3.6.14) through the auxiliary input. The recorded signal shall be reproduced and shall be impressed upon the input of a flutter discriminator which has a magnitude versus frequency characteristic that is uniform over a frequency range from 0.5 to 200 Hz to determine conformance with 3.6.17.

4.5.8.3 Channel isolation. A 100 and 5000 Hz signal at minus 20 dbm shall be applied to the auxiliary input and recorded at optimum record level on tracks 1 and 3. The recorded signals shall be reproduced on channel 3 and the channel 3 output level control shall be adjusted to give rated output level across the channel 3 output resistive load. The tape shall be reversed and tracks 2 and 4 of the same section of tape on which the 100 and 5000 Hz signals of tracks 1 and 3 were recorded shall be reproduced. The output level measured with the channel 3 output level control at the same level as when rated output level on tracks 1 and 3 was attained. The output levels of tracks 2 and 4 shall be related in db to the output level from tracks 1 and 3 to determine conformance with 3.6.16.

4.5.9 Regulation. The outputs of each channel shall be terminated with its proper resistive load (see 3.5.1.1). All tone controls shall be placed in the "FLAT" positions. A 1000 Hz signal of minus 40 dbm shall be applied to the microphone input. The microphone mode selector switch (see 3.4.3) shall be placed in the position for overriding all of the four channels simultaneously. Each channel output level control shall be adjusted to give rated output level across their respective output resistive loads. The output resistive loads of each channel shall be removed and the output voltage measured to determine conformance with 3.9.3. The test shall be repeated at input signal frequencies of 100, 4000, and 7500 Hz to determine conformance with 3.9.3.

4.5.10 Input and output impedance. The input and output impedances shall be measured for conformance with 3.4.4.1, 3.4.5.2, and 3.5.1.

4.5.11 Input level range. The input level range shall be measured to determine conformance with 3.4.4.1.

4.5.12 Level control linearity. The linearity of the level controls shall be measured to determine conformance with 3.11.4.

4.5.13 Mechanical requirements. The mechanical parameters shall be measured for conformance with 3.10.

4.5.14 Temperature and humidity. This test shall be conducted with all the equipment energized. The recorder-reproducer and the record player shall not be operated. The equipment shall be subjected to a temperature of plus 90°F. with a relative humidity of 90 percent for a period of 18 hours. The equipment shall then be immediately subjected to a temperature of plus 40°F. with a relative humidity of 90 percent for a period of 6 hours. The hot and cold cycle shall immediately begin again for a total of four cycles. Immediately after the fourth cycle, the equipment shall be given an operational test in accordance with 4.5.1 to determine that it is functioning properly. The equipment shall also be examined for corrosion and other damage. The following tests shall be made 24 hours after the fourth cycle to determine that the equipment complies with the respective requirements: distortion, frequency response, signal-to-noise, record player flutter, recorder-reproducer flutter.

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4.5.15 Accelerated life. All components of the equipment (including the record player and recorder-reproducer motors) shall be operated continuously for 250 hours. Once during each 24-hour interval the following shall be performed:

(a) Each of the two recorder-reproducers shall record and reproduce a 1000 Hz signal at 7.5 i.p.s. for 8 hours.
(b) The record player shall automatically reproduce 33-1/3 r.p.m. records for 7-1/2 hours and 45 r.p.m. records for 30 minutes.
(c) Each of two recorder-reproducers shall be continually switched from the fast forward operational mode to the rewind mode for one hour.
(d) The primary power to the equipment shall be switched on and off at a rate of 8 times per minute for five minutes.

The following tests shall be made after the life test to determine that the equipment complies with the respective requirements: operational, distortion, frequency response, signal-to-noise, record player flutter, recorder-reproducer flutter.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and sub-contractors shall be as specified in the individual order.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Audio entertainment system equipment.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. The onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation, packaging, packing and marking. The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C; packed level A or B, as specified and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. The equipment covered in this specification is intended to provide a centrally located audio entertainment system for Naval ships. The system's program material will be distributed by loudspeakers located throughout the ship.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Quantity of items required.
(d) Quantity of manuals required (see 3.3).
(e) Inventory control point for repair parts (see 3.3).
(f) Whether Federal stock number is to be entered on the identification plate (see 3.3).
(g) Preservation, packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 Preproduction. Invitations for bid should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5830-N016)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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